Band-1 Hadamard Order 7508 — Miyamoto (Paley-II Seed H_1876 q=937 + C_1878) + Paley-I IN SPK; Gap 2052 Still OPEN

H_1876 Paley-II PASS; C_1878 PASS; H_7508 Miyamoto PASS; H_7508 Paley-I PASS → 7508 IN SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7492_Miyamoto`; `2026-09-16_Band1_Hadamard_7444_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7508_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1876 and 7508 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1876} = Paley-II($q = 937$) and symmetric Paley conference C_{1878} over GF(1877) ($q = 1877 \equiv 1 \pmod 4$), **and** classical Paley-I H_{7508} = Paley-I($q = 7507$) (prime $\equiv 3 \pmod 4$). Classical SPK for 7508: Paley-I **PASS**es ($q = 7507$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL**s ($q = 3753 = 3^3 \cdot 139$ NOT prime power); Kronecker **FAIL**s (4×1877). Therefore order **7508 is IN SPK via Paley-I** (Miyamoto is an additional finite construction, not the SPK reason). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7492 Miyamoto-only NOT in SPK**. Prior SPK seeds **1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **1876** and **7508** folded into ONBOX/SPK when verified.

Abstract

Finite constructions of orders **1876** and **7508**.

- H_{1876} = Paley-II ($q = 937$) — preferred seed
- C_{1878} = Paley conference over GF(1877), $q = 1877$
- H_{7508} = Miyamoto(H_{1876}, C_{1878})
- Paley-I for 7508: **PASS** ($q = 7507$ prime $\equiv 3 \pmod 4$) — classical SPK
- Paley-II for 7508: **SKIP** ($q = 3753$ NOT prime power)
- Kronecker for 7508: **SKIP** ($4 \cdot 1877$ not admissible)

SPK: 7508 IN SPK via Paley-I q=7507. Seed 1876 enters SPK (Paley-II). Prior SPK **1872 / 1860 / 7444 / 1848 / 7396 / 1788 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7508.py`

Results. overall_pass=true; units 122/122; wall 709.2071s; all max|HH^T-nI|=0.

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Arithmetic

- Target $n = 7508 = 4 \cdot 1877$; 1877 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1876} and C_{1878} (Paley conference over prime field $q = 1877$).
- Seed: Paley-II $q = 937$ (prime $\equiv 1 \pmod 4$) → H_{1876} (Paley-I $q = 1875$ FAIL; Double FAIL; Kronecker 4×469 FAIL).
- SPK: Paley-I $q = 7507$ **PASS**; Paley-II $q = 3753$ **FAIL** (NOT prime power); Kronecker **FAIL → IN SPK via Paley-I**.
- Prior: 7492 Miyamoto-only NOT in SPK shipped; this ship $q = 1877 \rightarrow n = 7508$.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_1876 Paley-II	Paley-II q=937	0	3.5299
C_1878	Paley conf GF(1877) q=1877	0	4.6672
H_7508 Miyamoto	Miyamoto(H_1876 Paley-II, C_1878)	0	331.825
H_7508 Paley-I	Paley-I q=7507 DIRECT	0	360.8758
H_7508 Paley-II	SKIP q=3753=3^3 · 139	—	—
H_7508 Kronecker	SKIP 4 · 1877 not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **7492** lineage (**1872** Paley-I; **1860** Paley-II; **7444** Paley-II GF(61^2); **1848** Paley-I; **7396** Paley-II; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 7492/7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **1876** (Paley-II) + verified **7508** (Paley-I); **exclude** Miyamoto-only bonuses. Known count 1484. Next constructive gap **2052** (OPEN).

Quantity	Value
7508 in SPK	**True** (Paley-I q=7507)
1876 in SPK	**True** (Paley-II q=937)
1872 / 1860 / 7444 in SPK	**True** (prior)
7492 / 7156 / 7108 / 6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · finite constructions of 1876+7508 ONLY · SPK via Paley-I only (Miyamoto not SPK reason).

Repro

Lab: `/workspace/experiments/band1/hadamard_7508_miyamoto/`
Artifacts: `H1876_paleyII_pm1.npz`, `C1878_pm1.npz`, `H7508_miyamoto_pm1.npz`, `H7508_paleyI_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.